

CRITERIA SPECIFIC INDICATORS FOR DETERMINING SCORES

Attribute	Evidence examined
Functional suitability	Requirement-to-component coverage
Modularity	Responsibility separation, cohesion, coupling, boundaries
Maintainability	Dependencies, interfaces, separation of concerns, modification support
Scalability	Independence, extensibility, coupling, growth/scaling mechanisms
Interoperability	Interfaces, service boundaries, external integration
Security	Authentication, authorisation, access control, secure communication, data protection

Imp Note : If N/A is there in scores then there is no supporting evidence to justify either human/AI architectures

1) Functional Suitability

Score	Criterion-specific evidence
5	Architectural responsibilities/components provide clear and comprehensive support for the identified functional requirements, with no significant requirement left without an identifiable architectural responsibility.
4	The architecture provides strong support for most or nearly all functional requirements, with only minor omissions, ambiguities or weakly represented responsibilities.
3	The architecture provides meaningful support for the functional requirements, but several requirements are only partially represented, ambiguously allocated, or insufficiently addressed.
2	The architecture provides limited functional coverage, with substantial requirements lacking clear architectural responsibilities or requiring significant interpretation.
1	There is insufficient architectural evidence to demonstrate that the functional requirements are supported.

NOTE: I am determining ISO 25010 Sub Characteristics functional appropriateness , completeness, functional coverage as well so scores may be averaged based on 3 sub categories

Evidence I examine

- SRS functional requirements
- requirement-to-component mappings
- component responsibilities
- architectural diagrams
- service/module descriptions from published studies and ai generated descriptions

2. Modularity

Score	Criterion-specific evidence
5	Architecture demonstrates clear separation of responsibilities, cohesive components, well-defined boundaries and limited unnecessary coupling, with no significant structural weakness identified.
4	Architecture demonstrates good separation of responsibilities and component boundaries, with only minor coupling, overlap or boundary weaknesses.
3	Architecture demonstrates meaningful modular decomposition, but noticeable coupling, responsibility overlap or unclear boundaries remain.
2	Architecture demonstrates limited modular separation, with substantial coupling, poorly defined responsibilities or significant structural overlap.
1	Architecture provides little or no meaningful modular decomposition; responsibilities are substantially intertwined or architectural boundaries are not demonstrated.

Evidence I examine

- layers/modules/services
- component responsibilities
- dependency relationships
- coupling
- cohesion
- separation of concerns
- architectural boundaries

3. Maintainability

Score	Criterion-specific evidence
5	Architecture provides strong support for future modification, with clear component responsibilities, well-defined interfaces, limited unnecessary dependencies and effective separation of concerns.
4	Architecture provides good support for modification and evolution, with only minor dependencies, responsibility overlaps or structural limitations.
3	Architecture provides some support for maintainability, but noticeable dependencies, unclear responsibilities or limited separation may make future modification more difficult.
2	Architecture provides limited support for maintainability, with substantial dependencies, poorly separated responsibilities or significant modification difficulties evident from the design.
1	Architecture provides insufficient evidence of maintainable structure, with responsibilities and dependencies substantially inhibiting modification or evolution.

Evidence I examine

- separation of concerns
- dependency structure
- interfaces
- component boundaries
- business-logic separation
- extensibility mechanisms
- architectural patterns supporting change

4. Scalability

Score	Criterion-specific evidence
5	Architecture provides strong and explicit mechanisms for accommodating increased workload or system growth, with appropriate independence, extensibility, limited coupling and, where documented, resource/deployment scaling mechanisms.
4	Architecture provides good architectural support for growth and increased demand, although some scaling mechanisms or evidence remain limited or unspecified.
3	Architecture demonstrates some characteristics supporting scalability, but significant limitations exist in independence, coupling, extensibility or scaling mechanisms.
2	Architecture provides limited evidence of scalability, with substantial structural constraints that could make growth difficult.
1	Architecture provides no sufficient architectural evidence that it can accommodate increased workload or system growth.

Evidence I examine

- independent components/services
- extensibility
- coupling
- dependency structure
- horizontal/vertical scaling mechanisms if explicitly documented
- deployment architecture if available
- ability to add functionality without major redesign

Imp Note: I assessed scalability architecturally; I did not claim to empirically measure performance or load capacity.

5. Interoperability

Score	Criterion-specific evidence
5	Architecture provides clear and comprehensive mechanisms for external integration, including well-defined interfaces/service boundaries and appropriate integration mechanisms where required.
4	Architecture provides strong support for external integration, with only minor limitations in interface definition, integration coverage or supporting mechanisms.
3	Architecture provides some identifiable integration mechanisms, but significant gaps or ambiguities remain in external communication or interface design.
2	Architecture provides limited external integration support, with substantial gaps in interfaces, service boundaries or integration mechanisms.
1	Architecture provides insufficient evidence of mechanisms for interoperability with external systems or services.

Evidence I examine

- APIs
- REST/service interfaces
- gateways
- integration services
- external-system boundaries
- communication mechanisms
- data exchange mechanisms

6. Security

Score	Criterion-specific evidence
5	Architecture demonstrates comprehensive architectural consideration of relevant security concerns, including appropriate authentication, authorisation, access control, secure communication and/or data protection mechanisms where applicable.
4	Architecture demonstrates strong security support, but one or more relevant mechanisms are incompletely specified or minor limitations remain.
3	Architecture demonstrates some meaningful security mechanisms, but significant security concerns remain insufficiently addressed or specified.
2	Architecture demonstrates limited security consideration, with substantial gaps in relevant security mechanisms.
1	Architecture provides insufficient architectural evidence of security mechanisms or security considerations.

Evidence I examine

Whether the architecture incorporates mechanisms addressing:

- authentication;
- authorisation;
- secure communication;
- data protection;
- access control.